SPM 5708 Homework 4 (Correlations & Scatterplots)

This week, we discussed the topic of *sabermetrics* as it relates to the process of finding metrics that help us better evaluate the performances of players and teams. Metrics we would describe as "good" metrics are those that (a) accurately measure success/performance (relevancy), (b) differentiate between players of varying skill levels (discriminatory), and (c) separate skill from luck or other factors (consistency/stability). We also learned about a statistic, the *correlation coefficient,* that can be used to examine *the strength of the linear relationship between two quantitative variables,* and about the *scatterplots* that can be used to visualize them. This homework assignment further explores each of these elements. For questions that require an answer, please provide your answers in a different font color.

4.1 Exploring Advanced Batting Data Using Correlations and Scatterplots

1. Install and read the *baseballr* package into your current R session. This package provides useful functions for collecting and analyzing MLB data, including functionality for linking to Statcast, Baseball-Reference, and FanGraphs databases.

2. Let's use the following function to scrape a season's worth of batting data from the FanGraphs batting leaderboard:

   bat_data=fg_batter_leaders(startseason=2023,endseason=2023)

   If you get an error after running this, simply download the *bat_data* .csv file from Canvas and import it to R Studio before continuing with the next line of code.

3. To avoid potential outliers, filter the data to create a new data frame called *bat_use* that only contains players who made at least 50 plate appearances (*PA*) in this season.

4. Which of the following statistics displays the strongest correlation to infield hit percentage (*IFH_pct*), the percentage of batted ground balls that go for hits? Record each stat's correlation coefficient in relation to *IFH_pct* below:

   EV (exit velocity; avg. speed at which batted balls leave a player's bat)
   Spd (speed score; approximates a player's speed on a scale of 1 [slowest] to 10 [fastest])
   Contact_pct (contact percentage; the percentage of total swings that result in ball contact)
   GB_pct (groundball pct.; the percentage of a player's batted balls that are groundballs)

   Which metric displayed the *strongest* relationship with *IFH_pct*? How did you determine this?

   ANSWER:

   EV and IFG_pct = -0.1049752

   Spd and IFG_pct = 0.275508

   Contact_pct and IFG_pct = -0.01508206

   GB_pct and IFG_pct = 0.05089649

The metric that displayed the strongest relationship with IFH_pct was Spd (speed score). I determined this because the correlation coefficient is the highest from the other three interactions with a positive correlation coefficient of 0.275508.

5. Import the *Ohtani_2025* dataset and find the correlation between Ohtani's exit velocity or launch speed (*launch_speed*) and his launch angle (*launch_angle*) using this code:

   cor(Ohtani_2025$launch_speed,Ohtani_2025$launch_angle,use="complete.obs")

   Note that we must attach the *use="complete.obs"* function to the end of the code because there are NAs for some of the values in the data.

   Launch angle is recorded with a middle value of 0 to indicate a ball hit parallel to the ground; negative values imply a downward (groundball) trajectory, and positive values imply an upward (flyball) trajectory. Thus, it probably won't display a linear relationship with launch speed (exit velocity) since balls can be hit with pace at many angles. To see this, create a scatterplot with *launch_speed* on the x-axis and *launch_angle* on the y-axis. Don't forget to read in (via *library*) the *ggplot2* package prior to creating the plot.

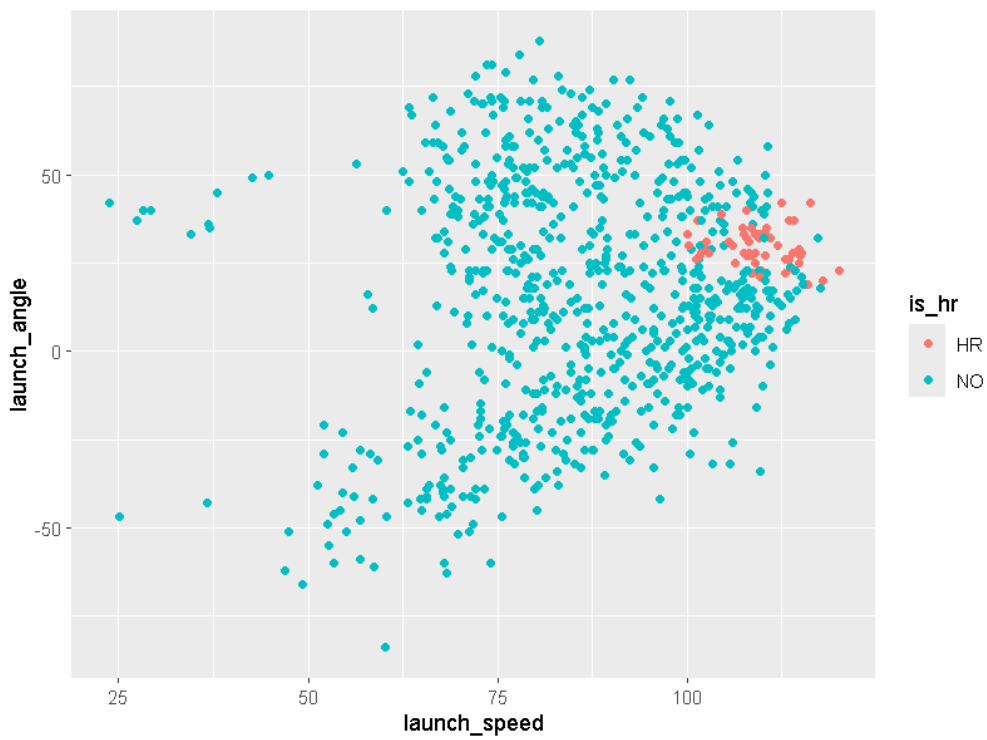
6. Run the following code to create a new column called *is_hr* that labels Ohtani's home runs:

   Ohtani_2025=Ohtani_2025%>%mutate(is_hr=ifelse(events=="home_run","HR","NO"))

7. Now, replace the last layer in your previous scatterplot code with this chunk of code and re-run it:

   +geom_point(aes(color=is_hr))

   What appears to be the "sweet spot" in terms of the launch angle and launch speed of Ohtani's home runs?



ANSWER: Apparently, the range for launch speed goes from 100 mph to 125 mph, and the range for launch angle goes from approximately 20 to 30 degrees.

8. Calculate the mean (average) launch angle and launch speed *of Ohtani's home run hits.* Report these values below.

   ANSWER: The mean launch angle of Ohtani's home run hits is approximately 29.62 degrees, and the mean launch speed of Ohtani's home run hits is approximately 109.16 mph.

4.2 Using Correlations and Scatterplots to Explore Basketball Data

1. Import the *NCAAB_Efficiency* dataset (.xlsx) and call it *NCAAB.* Start by calculating the correlation coefficients for points scored per game (*PPG*) and *WinPct*, and opponents' points per game (*OPPG*) and *WinPct*. Report them below.

   ANSWER: The correlation coefficient between PPG and WinPct is 0.6005331, and the correlation coefficient between OPPG and WinPct is -0.595546.

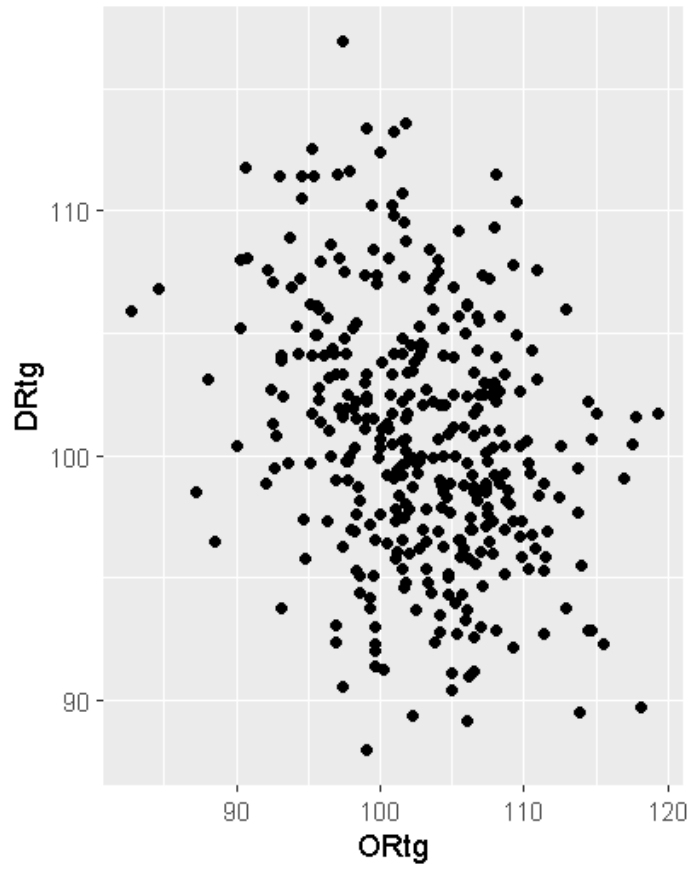
2. Now, find the correlations between *ORtg* and *WinPct*, and *DRtg* and *WinPct*. Comparing these results to the correlations from the prior question, what can you infer? Note that *ORtg* is a measure of points scored per 100 possessions, while *DRtg* measures points allowed (opponents' points scored) per 100 possessions.

   ANSWER: The correlation between ORtg and WinPct has a coefficient of 0.7736732 and the correlation between DRtg and WinPct has a coefficient of -0.7261376. Compared to the results above, we can assume that ORtg and DRtg have a stronger linear relationship (whether positive or negative) with WinPct than PPG and OPPG do.

3. Is there any relationship between being an efficient offensive team (*ORtg*) and an efficient defensive team (*DRtg*)? Report evidence in support of your answer.

   ANSWER: The correlation coefficient is -0.2924971 which tells us that there is a negative relationship and weak relationship between ORtg and DRtg.

4. Create a scatterplot that shows *ORtg* on the x-axis and *DRtg* on the y-axis.

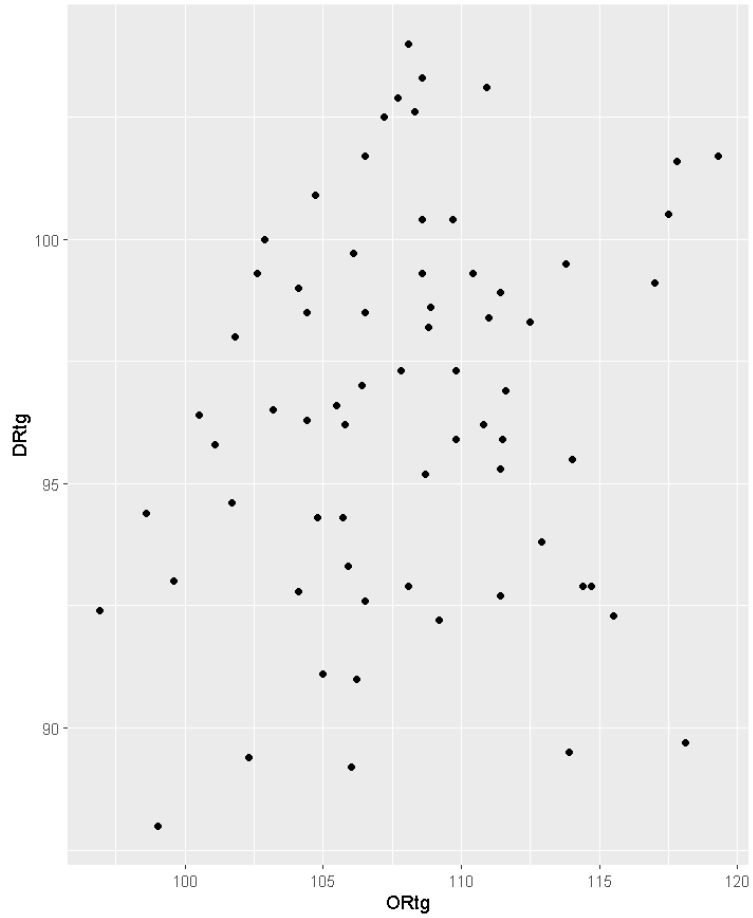


5. Next, find the correlation between the *Pace* (possessions per game) and *WinPct* (win percentage) variables. What does the resulting coefficient suggest?

   ANSWER: The correlation coefficient is -0.05719578 which tells us that there seems to be a very weak negative relationship between the variables. This tells us that the pace has close to no impact on Winning Percentage, which leads to think that other factors are more important when talking about Winning Percentage.

6. Does the correlation coefficient between *Pace* and *WinPct* change when we only examine tournament teams who have a 1 recorded in the *Tourney* column? In addition to finding the correlation coefficient, create a basic scatterplot of this relationship and paste it below.

   ANSWER: Yes, the correlation coefficient changes, it becomes positive but it is still a very weak relationship with a positive coefficient equal to 0.03327786. Below is the scatterplot.



4.3 Using Scatterplots to Examine the Linearity of Relationships

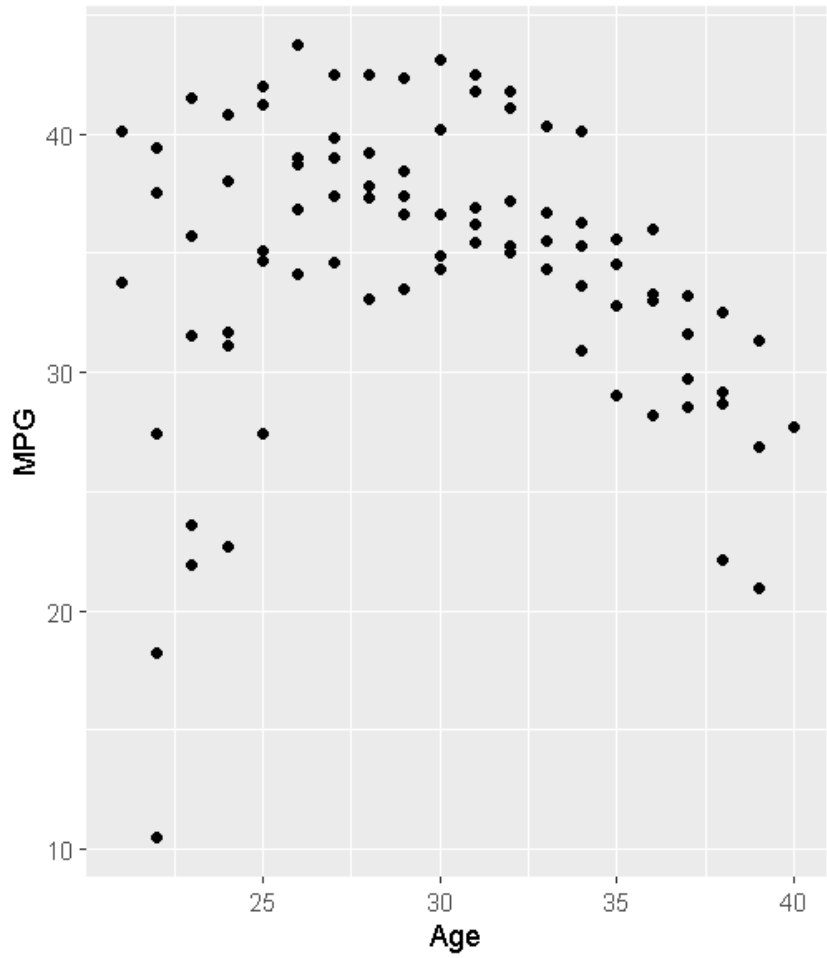
1. Import the *PGCareers* dataset (.csv) and note that each row corresponds to the season-level performance stats of a prominent (former) NBA point guard. All stats are recorded on a per-game basis aside from minutes played (*MP*). Start by creating a new variable column called *MPG* that shows minutes-per-game for the players in each row.

2. Would it be appropriate to run a correlation between players' *MPG* values and their *Age*? Why or why not? When you attempt this, what value do you obtain?

   ANSWER: No, because the relationship is no linear, when players are young and when they are reaching their retirement age, they play less, while on the other hand, after some years of experience, they start to play more. The correlation coefficient obtained is -0.126942.

3. Perhaps a scatterplot could help us visualize the shape of the trend between players' minutes-per-game values and their age. Create a scatterplot using the following code and paste it below:

   ```
   ggplot(PGCareers,aes(x=Age,y=MPG))+geom_point()
   ```

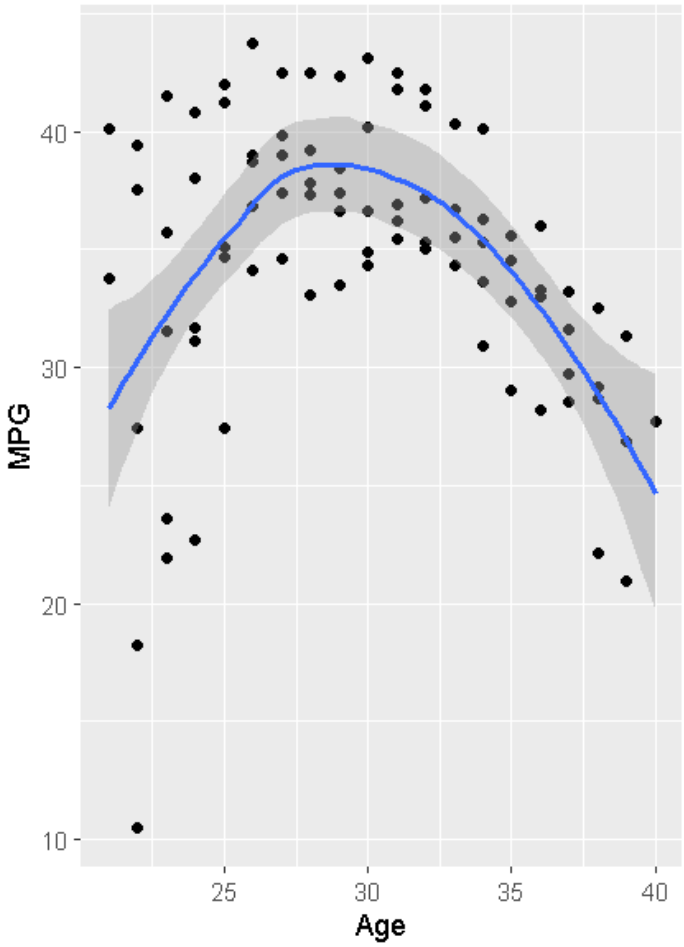


4. Note how we can add the following layer to our *ggplot* code to display a “line of best fit” that better portrays the nonlinearity of the relationship:

   ggplot(PGCareers,aes(x=Age,y=MPG))+geom_point()+geom_smooth(method=”loess”)

   What does the shape of this trendline suggest about the relationship between playing time and age in basketball?

   ANSWER: The relationship is non-linear and it seems that when players are young and reaching retirement age, they tend to play less minutes per game.



Submit your completed R Script and answers to the requested questions in this Word document to the Canvas drop box by the next class.